Python Data Cleansing Pipeline (ETL)

Technical Validation & Impact Report — Q2 2026

Prepared by: Keisy Fernanda

Role: Data Analyst / Analytics Engineer

Stack: Python (Pandas, NumPy)

📌 PROJECT GOAL

The business problem was the integration of a legacy database that exported heavily corrupted operational CSV files. Manual cleaning in Excel was consuming excessive hours and introducing critical human errors. The goal was to write a robust, automated Python script using the Pandas library to execute an ETL (Extract, Transform, Load) pipeline, ensuring **100% data conformity** for corporate reporting.

🛠️ TECHNICAL PROCESS & VALIDATIONS

The script implements several critical cleansing and standardization steps:

- **Vectorized Text Normalization:** Automated trimming of white spaces and casing standardization (Title Case for names, UPPER CASE for countries), removing common data entry errors.
- **Intelligent Missing Value Imputation:** Revenue fields with null values are automatically identified and populated using the dataset's **median revenue value**, preserving statistical integrity for financial dashboards.
- **Business Logic Guardrails:** Anomalies, such as negative purchase quantities, are programmatically corrected, defaulting them to **1** to prevent dashboard visual errors without halting the pipeline.
- **Datetime Standardization:** A robust conversion of various corporate date formats into a single, standardized format, using forward-filling to handle missing dates in chronological order.

📊 OPERATIONAL & BUSINESS IMPACT

Deployment of this automated Python solution yields immediate results:

1. **Efficiency Gain:** Eliminates manual data preparation time, cutting weekly administrative effort from hours to just a few seconds.
2. **Data Integrity:** Guarantees structural and content conformity, preventing downstream dashboard breaks and ensuring reliable financial reporting.
3. **Scalability:** The script can process large-scale datasets beyond traditional spreadsheet capabilities.